Research Methodology

TU Delft

Research methodology content

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Written with extensive use of Claude LLM

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Contents

How to Use This Guide

This guide is built for the open-book exam, so it is engineered for fast retrieval rather than linear reading. It is split into two halves. The first half, the Overview at a Glance, is a compact map of the whole course. The second half, the numbered week chapters, is the detailed reference. The same four layers run through both halves, but they are deliberately separated so that scanning and deep reading do not get in each other's way.

The four layers

The first layer is the **snapshot card** at the head of each week. It carries the weekly connection diagram, four to six exam-usable takeaways, and a Connects from and Connects to line that places the week inside the wider course flow.

The second layer is the **part summary**, one coloured box per lecture part. It states what the part establishes, why it matters, and where its full detail lives. It is the filter that answers whether the detail underneath is needed at all.

The third layer is the **bullet content** under each *Content* heading. It reproduces the lecture logic faithfully, listing definitions, classifications, criteria, and the worked examples given in the slides. Terms defined for the first time are set in bold and registered in the index at the back.

The fourth layer is the **prose explanation** under each *Explanation* heading. It turns the bullets into understanding by expanding the reasoning, the nuance, and the links between concepts.

The two halves

The Overview at a Glance collects layers one and two for all six weeks back to back, with no detail in between. Read it to get the entire shape of the course in a few pages, or to locate which week and part a question belongs to. Each part summary there ends with a link to the exact section in the second half where its content lives.

The numbered week chapters hold layers three and four, the bullets and the prose, organised by part. Each chapter opens with a page reference back up to its overview entry, so navigation works in both directions. The cross-week threads, the report guide, and the appendices follow the week chapters.

The colour scheme

The snapshot cards and headings follow the colours of the course connection diagrams. Green marks the conceptual track, the chain that runs from the language of research through conceptual design to the written report. Red marks the empirical track, the chain that runs from sampling and data through validation to data management. Blue-grey marks the integrative topics that draw both tracks together. The course map on the next page shows the two tracks in full.

Finding things on paper

Every internal pointer uses `\autoref`, which prints the entity type with the number, so a reference reads “Section 2.3” or “Appendix A” rather than a bare number. This makes each pointer usable directly on a printed page. The fastest lookup tool is the index at the very back, where every bold term is listed with its page number. The glossary in Appendix B gives one-line definitions with a pointer to the section where each term is treated in full.

Course Map

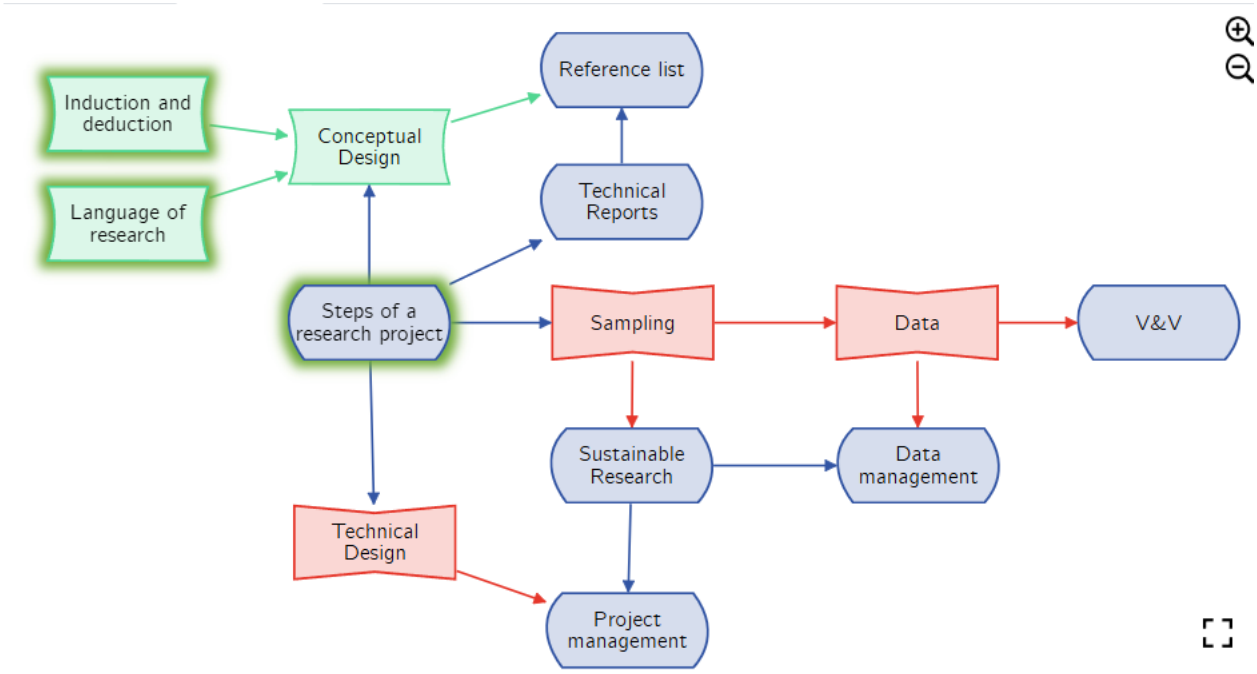


Fig. 1: The course as two parallel tracks. Green nodes form the conceptual track, red nodes the empirical track, and blue-grey nodes the integrative topics that join them.

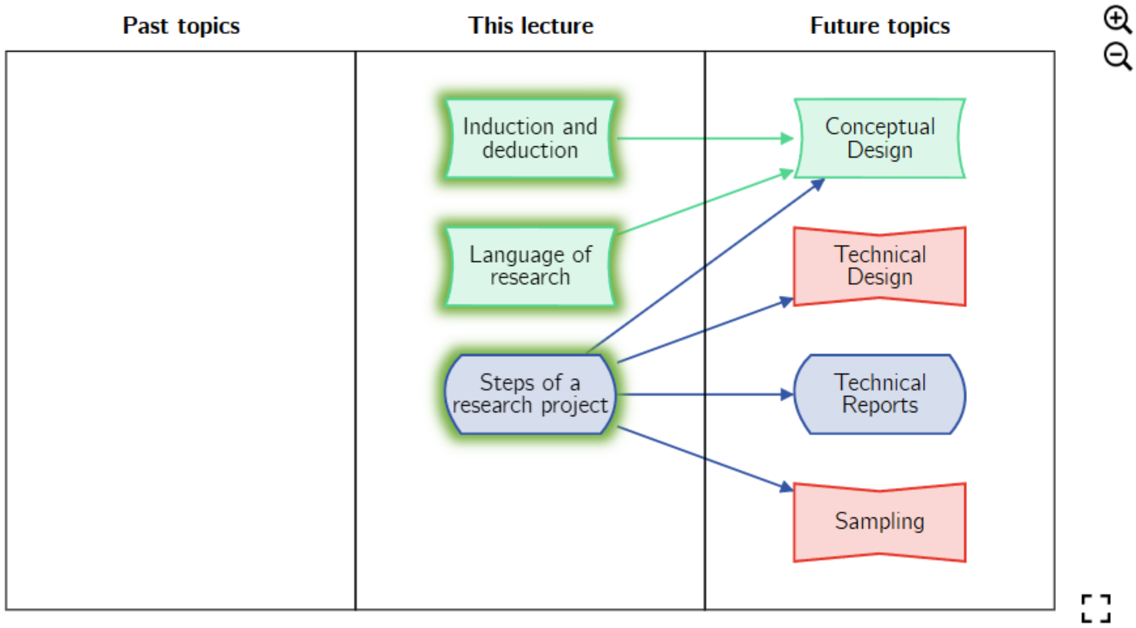
Figure 1 lays out the whole course as two parallel tracks that both start from Week 1. The conceptual track runs in green from the language of research and the induction and deduction logic of Week 1, into the conceptual design of Week 2, and finally into the technical report and reference list of Week 6. It is concerned with what is studied and how the argument is framed. The empirical track runs in red from the steps of a research project in Week 1, into sampling and data in Week 3, into verification, validation, and sustainable research in Week 4, and into data management and project planning in Week 5. It is concerned with how evidence is gathered, checked, and managed. The single node “Steps of a research project” sits at the centre because it feeds both tracks at once, branching into conceptual design, technical design, the report, and sampling.

The two integrative bridges are Sustainable Research and Data Management. Sustainable research grows out of sampling on the empirical side yet feeds project management and data management, which forces the resource and reproducibility questions of one track onto the planning decisions of the other. Data management collects the output of data, sustainable research, and technical design, so it is where the two tracks finally meet. Reading the guide week by week walks down both tracks in step, and the snapshot card at the head of each chapter names exactly which upstream nodes feed the week and which downstream nodes depend on it.

Overview at a Glance

This half of the guide is the fast map. It collects the snapshot card and the part summaries for all six weeks, with no detailed content in between, so the whole shape of the course can be scanned in a few pages. Each part summary ends with a link to the section in the detailed reference half where its bullets and prose live. The detailed chapters in turn link back to the matching week here.

Week 1 — Introduction to Research



Week 1 — Introduction to Research

Connects from: Course start, no prerequisites → **Connects to:** Conceptual and technical design (chapter 2), sampling (chapter 3), reports (chapter 6)

- The research process runs in six fixed steps: the problem, the sample, the design, measure, analyse, conclude.
- A hypothesis is a concrete prediction, and its form fixes whether the study is descriptive, relational, or causal.
- The variable is the smallest unit of research language, and its values are attributes that must be exhaustive and mutually exclusive.
- An observed relationship may be correlational, causal, or produced by a third variable, so correlation never proves causation.
- Deduction reasons from general to specific and induction from specific to general, and the hourglass joins the two in one study.
- Feasibility rather than interest decides whether an idea becomes a project, and the literature review sets that idea in context.

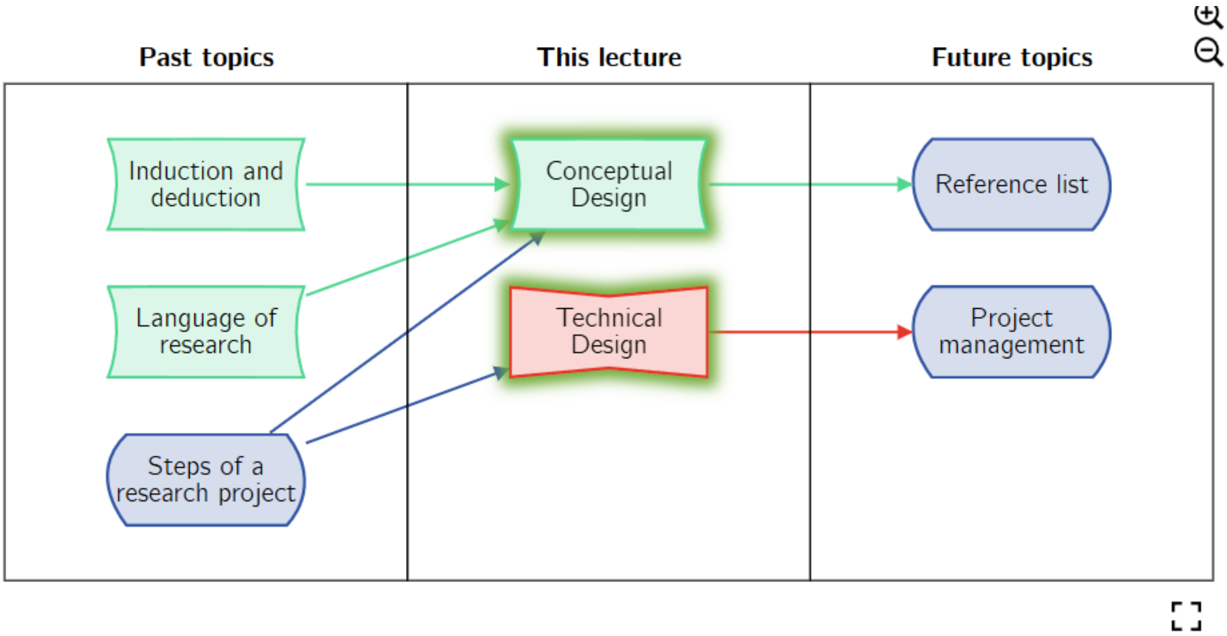
Part 1 — The Research Process. This part fixes the skeleton that every later week hangs on: the six-step pipeline from a broad problem to a defensible conclusion. It matters because each step constrains the next, so a weak sample or an unstated limitation propagates all the way to the conclusion. The design step is the hinge that Week 2 expands into conceptual and technical design (chapter 2), and the measure step is what Week 3 turns into sampling and data (chapter 3). Full detail in section 1.1.

Part 2 — The Language of Research. This part supplies the vocabulary that makes the rest of the course precise: hypothesis, study type, variable, attribute, and the kinds of relationship between variables. It matters because the form of the hypothesis decides the type of study, and the type of study decides what evidence will count. The closing warning, that correlation is not causation, is the single most reused idea in the guide and reappears in validation (section 4.1). Full detail in section 1.2.

Part 3 — The Hourglass: Deduction and Induction. This part gives the shape of the reasoning that runs through a single study, narrowing from broad questions down to measurement and widening back out to general claims. It matters because the two directions of inference, deduction downward and induction upward, carry different risks, and induction is where most flawed conclusions are born. The operationalise step at the neck of the hourglass is exactly what Week 2 formalises (section 2.2). Full detail in section 1.3.

Part 4 — Where Research Ideas Come From. This part covers the front end of a project, how ideas arise, whether they are feasible, and how a literature review frames them. It matters because an interesting idea that fails on time, ethics, cost, or access never becomes a viable project, so feasibility is a gate rather than an afterthought. The literature review introduced here is what locates the gap that Week 2 turns into a research objective (section 2.2). Full detail in section 1.4.

Week 2 — Research Design



Week 2 — Research Design

Connects from: Language of research, induction and deduction, steps of a research project (chapter 1) → **Connects to:** Reference list (chapter 6), project management (chapter 5)

- Research design splits into conceptual design, which fixes what, why, and how much, and technical design, which fixes how, where, and when.
- Conceptual design has four phases: the objective, the research framework, the research question or conceptual model, and definition with operationalisation.
- A good research objective is informative, useful, feasible, and clear, and it names a contribution and the way that contribution is delivered.
- Research questions must be open-ended, number at least two, and be relevant, researchable, anchored, and precise.
- Technical design has three phases: research strategy, research material, and research planning.
- Reproducibility and the code of conduct govern conduct, and any work with human subjects requires HREC approval.

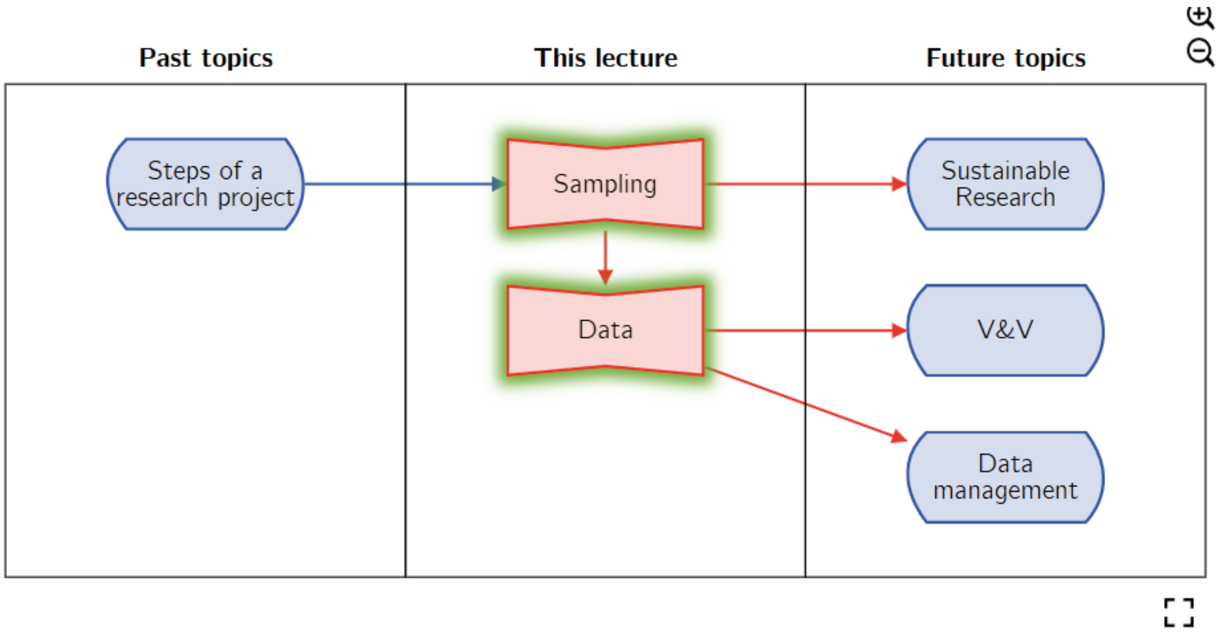
Part 1 — Strategic Areas. This part splits the single design step of Week 1 (section 1.1) into two strategic areas that are tackled in order. Conceptual design models the content of the work and answers what, why, and how much, while technical design plans the execution and answers how, where, and when. The split matters because the technical decisions are meaningless until the conceptual ones are settled, so the order is a dependency rather than a preference. Full detail in section 2.1.

Part 2 — Conceptual Design. This part is the engine room of conceptual design, where the objective, the research questions, the framework, and the operationalisation are built. It matters because these four outputs are what the technical design consumes, and a weak research question set is the usual reason a technical design will not come together. The criteria and rules given here are the same ones collected as checklists in Appendix A. Full detail in section 2.2.

Part 3 — Technical Design. This part turns the operationalised concepts into an executable plan through three phases, the research strategy, the research materials, and the research planning. It matters because this is where feasibility stops being abstract and becomes a schedule with milestones, equipment, and contingencies. The strategy criteria here echo the feasibility gate of Week 1 (section 1.4), and the planning tools named here are developed fully in Week 5 (section 5.2). Full detail in section 2.3.

Part 4 — Ethics and Conduct. This part sets the ethical baseline for the whole course, built on reproducibility, traceability, proper credit, and the code of conduct. It matters because these duties constrain every later choice about data and method, and because human-subject work is gated by a formal committee rather than left to judgement. The traceability requirement here is the same one reproducibility and data management depend on later (section 4.2, section 5.1). Full detail in section 2.4.

Week 3 — Research Material and Data



Week 3 — Research Material and Data

Connects from: Steps of a research project (chapter 1) → **Connects to:** Verification and validation, sustainable research (chapter 4), data management (chapter 5)

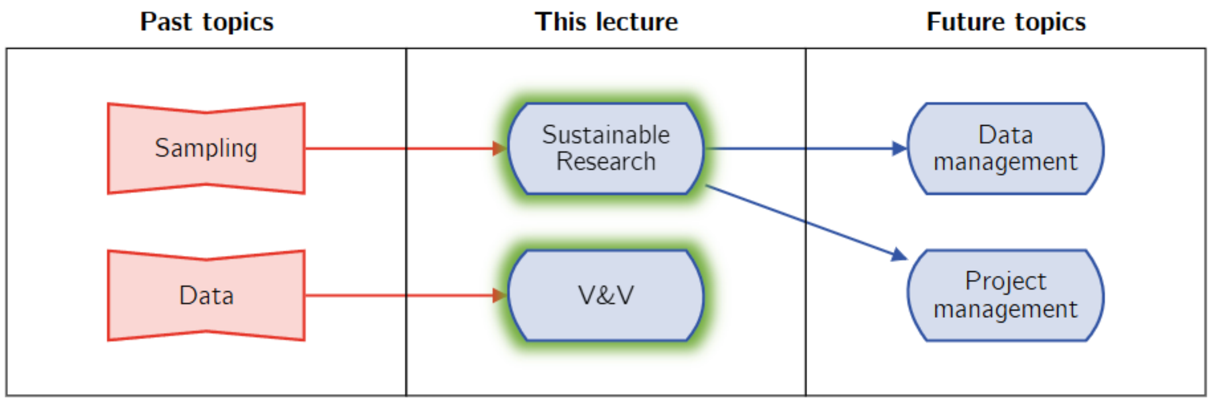
- The population is the entire collection in the field, and the sample is the representative subset that is actually studied.
- The sampling method narrows from the theoretical population, through the study population and the sampling frame, to the sample.
- Probability sampling supports generalisation to the population, while non-probability sampling never can and must say so explicitly.
- The standard error measures the precision of a sample, falling as the sample size grows and as the standard deviation shrinks.
- Data is qualitative or quantitative, and its type follows from the strategy, the materials, and the defined variables.
- Data quality must be checked before analysis, and a detailed research logbook is what makes the whole project traceable.

Part 1 — Population, Sample, and the Sampling Method. This part makes the sample step of Week 1 (section 1.1) precise by separating the population from the sample and showing how one narrows to the other. It matters because representativeness is what licenses any statement about the population, and the inductive leap from sample to population is exactly the risk flagged in the hourglass (section 1.3). The four-stage funnel given here names where access and listing constraints quietly shrink the reachable population. Full detail in section 3.1.

Part 2 — Types of Samples and Standard Error. This part classifies sampling into probability and non-probability methods and introduces the standard error as the measure of a sample's precision. It matters because the probability boundary is the same boundary as generalisability, so the sampling choice decides in advance what the study can claim. The standard error then quantifies how tightly the sample estimates the population, with two levers that reappear in the resource questions of Week 4 (section 4.2). Full detail in section 3.2.

Part 3 — Data Sources, Quality, and the Logbook. This part covers where data comes from, the legal categories that constrain it, the qualitative and quantitative split, and the quality checks and logbook that protect it. It matters because the data rules are hard gates, GDPR for personal data and a knowledge embargo for protected fields, that can stop a project before it starts. The logbook introduced here is the practical instrument behind the traceability demanded by ethics (section 2.4) and the full data management plan of Week 5 (section 5.1). Full detail in section 3.3.

Week 4 — Validation and Sustainability



Week 4 — Validation and Sustainability

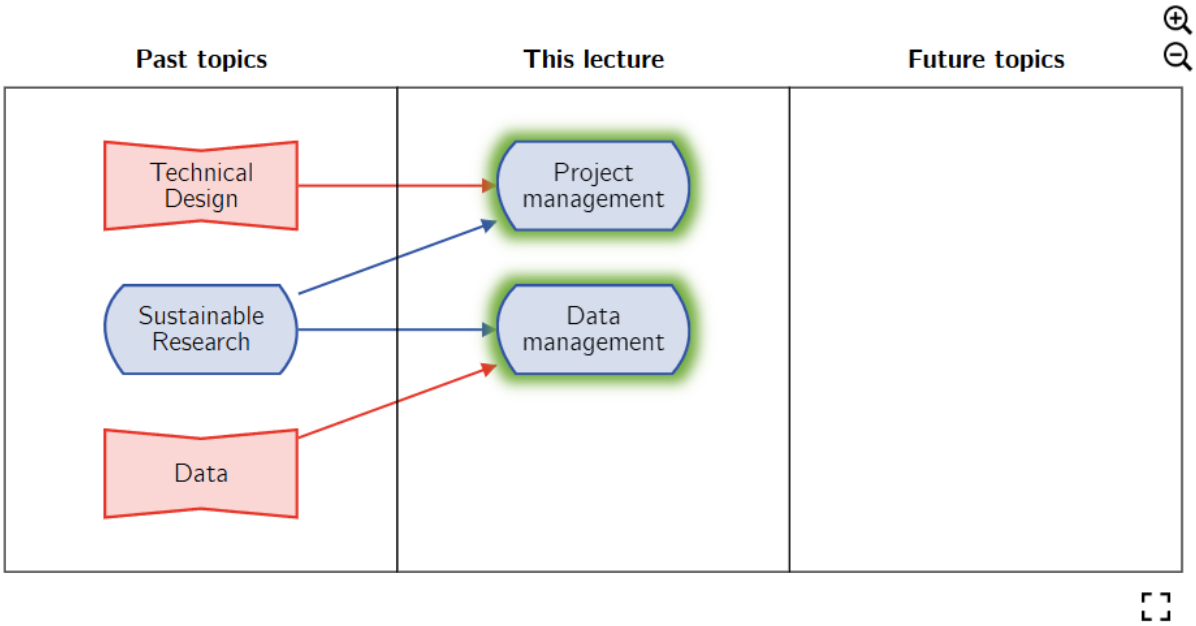
Connects from: Sampling and data (chapter 3) → **Connects to:** Data management and project management (chapter 5)

- Verification proves the method was used correctly and suits the topic, while validation proves the outcomes are true on strong evidence.
- Both are needed for credibility, professional use, and reproducibility, and both must be designed into the plan with time reserved.
- In industry, verification is internal quality control against specifications and validation is external quality assurance of fitness for purpose.
- The Engineering for One Planet framework weighs research on its value, its reproducibility, and its resource management.
- Reproducibility depends on good data management and is blocked by complexity, technology change, human error, and intellectual property.
- Resource management spans digital habits and methodology choices, each judged on whether the cost matches the accuracy actually needed.

Part 1 — Verification and Validation. This part draws the line between proving a method was used correctly and proving the result is true, the two distinct guarantees a credible study must give. It matters because a method can be executed flawlessly and still yield a false conclusion, which is the correlation-and-causation problem of Week 1 wearing a different name (section 1.2). Both checks consume time and must be planned for, linking straight back to research planning (section 2.3). Full detail in section 4.1.

Part 2 — Engineering for One Planet. This part introduces the Engineering for One Planet framework, which judges research on its value, its reproducibility, and its resource use. It matters because it reframes sustainability as a design constraint that touches topic choice, documentation, and method, rather than an ethical extra bolted on at the end. The reproducibility axis depends on the data management of Week 5 (section 5.1), and the resource axis gives weight to the sample-size and accuracy trade-off from Week 3 (section 3.2). Full detail in section 4.2.

Week 5 — Data Management and Project Planning



Week 5 — Data Management and Project Planning

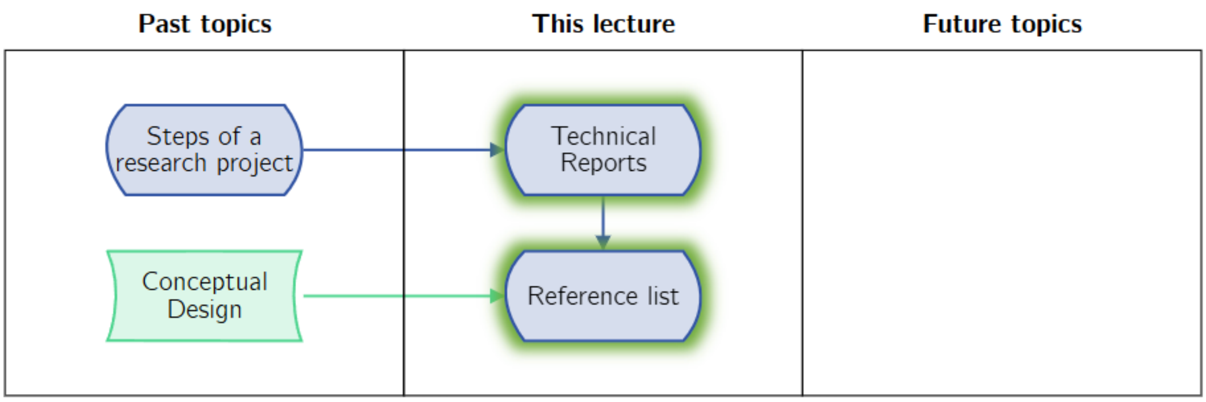
Connects from: Technical design (chapter 2), sustainable research and data (chapter 3, chapter 4) → **Connects to:** Reporting (chapter 6)

- A data management plan has six sections, from data description to responsibilities, and must be designed before any data is collected.
- FAIR data is findable, accessible, interoperable, and reusable, which is the goal that good documentation and metadata serve.
- The 3-2-1 rule keeps three copies on two media with one off-site, fixing the backup strategy from the start.
- Project management gives a research project structured timelines, optimised resources, early risk detection, and clear communication.
- A Work Breakdown Structure divides the project into work packages and tasks down to a level where each task can be estimated.
- A Gantt chart schedules the work and exposes the critical path, the chain that sets the shortest possible project duration.

Part 1 — The Data Management Plan. This part turns the logbook habit of Week 3 (section 3.3) into a designed system, the six-section data management plan. It matters because the plan must exist before data collection, since data cannot be organised retrospectively, and because it is the concrete form of the reproducibility duty from Week 4 (section 4.2). The plan is not a deliverable for this course, but its structure is examinable. Full detail in section 5.1.

Part 2 — Project Management in Research. This part develops the planning tools named in Week 2 (section 2.3) into full project management, from task estimation through the Work Breakdown Structure to the Gantt chart. It matters because a schedule is only trustworthy once the work is decomposed to estimable pieces and the critical path is known. The same precondition recurs here as in Week 2, that planning is possible only after both designs are settled. Full detail in section 5.2.

Week 6 — Reporting and Writing



Week 6 — Reporting and Writing

Connects from: Steps of a research project (chapter 1), conceptual design (chapter 2) → **Connects to:** The Research Methodologies report (chapter 8)

- A technical report conveys specific information about a technical subject to a specific audience for a specific and practical purpose.
- The Research Methodologies report conveys a research plan to a technical peer in order to organise the research activities.
- Every report has a head that introduces, a main body of connected sections, and feet that conclude without adding new information.
- Choose one reference style and keep it, either author-date such as APA or numbered such as AIAA, each with its own trade-off.
- A reference list should use at least three document types, lean on peer-reviewed articles, and stay mostly under ten years old.
- Reader-centred writing guides the reader in through titles, figures, and conclusions first, and a final checklist closes the report.

Part 1 — Technical Reports. This part defines what a technical report is and maps the generic report skeleton onto the Research Methodologies report specifically. It matters because the report is where the conceptual and empirical tracks of the whole course are finally written down, and its structure mirrors the work done in Weeks 2 and 5. The section-by-section mapping is developed in full in the report guide (chapter 8). Full detail in section 6.1.

Part 2 — Referencing. This part sets the rules for sourcing and for the two accepted citation styles, author-date and numbered. It matters because references are where the give-credit duty of Week 2 ethics (section 2.4) becomes a concrete, checkable practice, and because mixing styles is a common avoidable error. The two styles trade in-text information against readability in opposite ways. Full detail in section 6.2.

Part 3 — Writing Style. This part collects the style guidance, from defining the target reader, through graph conventions and the responsible use of AI tools, to the final writing checklist. It matters because reader-centred choices about titles, figures, and layout are what get a report read at all, and the figure conventions here extend the data presentation of Week 3. The closing checklist is the last gate before a report is considered finished. Full detail in section 6.3.

Chapter 1

Week 1 — Introduction to Research

Snapshot and part summaries: see the overview on page 4.

1.1 Part 1 — The Research Process

1.1.1 Content

- The process divides into six steps: **the problem**, **the sample**, the **research design**, **measure**, **analyse**, and **conclude**.
- The problem is a deliberately broad formulation, motivated by societal relevance and by the pursuit of fundamental knowledge.
- The problem is too broad to tackle whole, so it is split into portions or samples, each expected to contribute to the question.
- Choosing the right sample drives the success of the work, and knowing the limitations of that sample is equally important.
- Research design is the glue of the project, a chosen method that clarifies what will be measured and why it is measured.
- Measurement is the observing and recording of observations through experiments, analytical calculations, numerical simulations, or questionnaires and focus groups.
- Analysis proceeds in three steps, data processing, data grouping, and data analysis, because measured data alone answers nothing.
- The conclusion states results and implications, compares them to expectations, and reports limitations, next steps, and relevance.

1.1.2 Explanation

The six steps are a pipeline in which the freedom available at each stage is set by the choices made before it. The problem is stated broadly on purpose, but breadth is also why it cannot be attacked directly, which is what forces the move to a sample. The sample is therefore the pivotal decision of the whole project, because representativeness and known limitations are exactly what bound the claims the final conclusion is allowed to make. A sample chosen for convenience rather than representativeness does not invalidate the work, but it does cap the conclusion at the studied portion rather than the full problem.

The design step is called the glue because it is where the abstract problem becomes an executable procedure, and it is the step Week 2 splits into the conceptual question of what and why and the technical question of how and when (section 2.1). Measurement then commits the project to specific instruments, and the four named forms map directly onto the data types treated in Week 3 (section 3.3). Analysis is separated from measurement to make the point that raw observations are not yet evidence, since processing and grouping must precede interpretation. The conclusion deliberately returns to the same two motivations that opened the problem, societal relevance and fundamental knowledge, so that the project is judged against the question it set itself rather than against the data it happened to collect.

1.2 Part 2 — The Language of Research

1.2.1 Content

- A **hypothesis** is a specific statement of prediction that describes in concrete terms what the study expects to happen.
- A **descriptive study** documents the values of an unknown variable, such as the proportion of votes between two parties.
- A **relational study** investigates the relationship between two or more variables, such as age against voting proportion.
- A **causal study** determines whether one or more variables cause or affect one or more outcome variables.
- A **variable** is the smallest block in the language of research, and the values it can take are its **attributes**.
- Attributes are quantitative or qualitative, and they must be exhaustive over all possible answers and mutually exclusive.
- Qualitative values can rest on quantitative judgements, and all qualitative data can be summarised and manipulated numerically.
- The pattern of a relationship is none, positive, negative, or curvilinear, and its nature is correlational, causal, or a third-variable problem.
- **Correlation is not causation**, so an association alone never establishes that one variable drives another.

1.2.2 Explanation

The hypothesis is placed first because it is the lever that selects the study type, and the three study types form a ladder of ambition. A descriptive study only fixes the value of something unknown, a relational study asks whether two such values move together, and a causal study makes the much stronger claim that one of them drives the other. Each rung demands more from the design and the data than the one below it, which is why naming the type early prevents a project from promising a causal answer on relational evidence. The variable and its attributes are the atoms beneath all three, and the requirements that attributes be exhaustive and mutually exclusive are what make a measurement well defined rather than ambiguous.

The distinction between the pattern and the nature of a relationship is the part most worth holding onto. The pattern is the observable shape of the association, which can be read straight off the data as none, positive, negative, or curvilinear. The nature is the interpretation placed on that shape, and here the third-variable problem sits between honest correlation and a genuine causal link. A strong positive pattern is fully compatible with there being no causal arrow at all, because an unmeasured third variable can drive both observed quantities at once. This is the precise sense in which correlation is not causation, and it is the reason a causal study needs a design that rules out the third variable rather than merely measuring the association.

1.3 Part 3 — The Hourglass: Deduction and Induction

1.3.1 Content

- The **hourglass model** begins with broad questions, narrows and focuses, then operationalises before observing.
- At the neck it measures and analyses data, then widens again to reach conclusions and generalise back to the questions.
- **Deductive reasoning** moves from the more general to the more specific.
- **Inductive reasoning** moves from specific observations to broader generalisations and theories.
- Induction is fallible, since a generalisation drawn from limited observations can be plainly false even when each observation is true.
- Worked cautionary cases include inferring that one third of all students like apples from one third of a single sample, or that all planets hold water from Earth and Mars.

1.3.2 Explanation

The hourglass is a single picture for the whole inferential journey of a study, and its two cones are the two directions of reasoning. Coming down the upper cone is deductive, since the project moves from a general question to the specific operationalised quantities it will actually observe. Going up the lower cone is inductive, since the project moves from the specific measurements back to a general claim about the original question. The neck is where operationalisation and measurement live, and it is narrow on purpose, because the whole weight of the later generalisation rests on the few quantities chosen there.

Induction is singled out as the dangerous direction because it is not truth-preserving in the way deduction is. Every cautionary example shares the same defect, a leap from a sample or a handful of cases to a universal statement that the evidence does not support. The apple example fails because one sample need not match the population, which is precisely the representativeness concern that Week 3 makes statistical (section 3.1). The planets example fails because two confirming instances cannot establish a universal law. Recognising these as inductive overreach is the same skill as recognising that correlation is not causation, and both are guards on the widening half of the hourglass where conclusions are made.

1.4 Part 4 — Where Research Ideas Come From

1.4.1 Content

- Research ideas come from practical problems in the field, from a lack of specific knowledge, and from the conclusions of other studies.
- **Feasibility** is a trade-off between rigour and practicalities, and it gates whether an idea can become a project.
- The practicalities are duration, ethical considerations, required cooperation, manageable cost, and access to the tools needed.
- A **literature review** identifies related research and sets the project within a conceptual and theoretical context.
- A good review answers what is already known and in which discipline, and which theories and concepts are used and how they are defined.
- It also answers how the topic is researched and which methods exist, and which questions remain unanswered.
- Search recommendations: concentrate on credible scientific literature and identify the keywords of the topic for academic search engines.
- Build a mind map linking authors, institutions, and areas, and from a good baseline article follow both its references and its citations.

1.4.2 Explanation

The three sources of ideas share a structure worth noticing, since each is a gap of a different kind. A practical field problem is a gap between what is needed and what exists, a lack of specific knowledge is a gap in the literature itself, and the conclusions of other studies are explicit signposts to the next gap. Feasibility then acts as the filter between an idea and a project, and it is framed as a trade-off rather than a checklist because rigour and practicality genuinely pull against each other. A more rigorous design usually costs more time, more cooperation, or more equipment, so the feasibility questions are the points where that tension is made explicit and resolved.

The literature review is presented as the tool that does two jobs at once, locating the gap and framing the contribution that will fill it. The four questions a good review answers move from the state of knowledge, through the theories and definitions in use, to the methods available and finally to the open questions, which is the same order a research objective is later built in. The search advice is practical rather than abstract, since following the references of a strong baseline article walks backward through the intellectual lineage of a topic, and following its citations walks forward to current work. This is why a single well chosen article can seed an entire review, and it is the habit that Week 6 relies on when it asks for a reference list of sufficient breadth and currency (section 6.2).

Chapter 2

Week 2 — Research Design

Snapshot and part summaries: see the overview on page 6.

2.1 Part 1 — Strategic Areas: Conceptual and Technical Design

2.1.1 Content

- Research design has two strategic areas, **conceptual research design** and **technical research design**.
- Conceptual design models the content of the research and determines what, why, and how much is going to be studied.
- Conceptual design has four phases: the objective, the research framework, the research question and conceptual model, and definition and operationalisation.
- Technical design is the set of activities needed to realise the objectives set in the conceptual phase, and it concerns how, where, and when.
- Technical design has three phases: research strategy and research material, both answering how, and research planning, answering where and when.

2.1.2 Explanation

The two strategic areas are a clean division between deciding and doing, and the course is deliberate about which comes first. Conceptual design is the content layer, where the project commits to what it will study, why that is worth studying, and how much of it is in scope. Technical design is the execution layer, where the project commits to the instruments, the location, and the schedule. The reason conceptual must precede technical is that every technical choice is an answer to a conceptual question, so a method chosen before the objective is fixed is a method chosen for no stated purpose.

The phase counts are worth memorising because they organise the rest of the week. The four conceptual phases move from the objective, through the framework and the research question, to the operationalisation that hands concrete variables to the technical side. The three technical phases then take those variables and decide how to obtain them, with what materials, and on what timeline. The planning phase reaches forward into Week 5, where the same systems-engineering tools become full project management (section 5.2), so this part is the seam between research design and project execution.

2.2 Part 2 — Conceptual Design: Objectives, Questions, and Operationalisation

2.2.1 Content

- **Theory-oriented objectives** are theory development, building a theory about something not yet known, and theory validation, testing whether a proposed variable relationship exists.
- **Practice-oriented objectives** are problem analysis, diagnosis, design, and change.
- Problem analysis recommends what technological developments solve a problem, and diagnosis recommends how to improve or optimise a current situation.
- Design produces a solution to a technical problem, and change alters a practical situation and analyses the impact.
- The **research objective** follows the template: the objective is (a) the contribution to the solution, by (b) the way that contribution is provided.
- **Criteria for a good research objective**: informative, useful, feasible, and clear.
- **Research questions** yield information useful and necessary for the objective and enable the technical design to be built.
- Formulate a question by naming the key concept, thinking qualitatively first, then linking to quantitative evidence and a testable main hypothesis.
- Pitfalls: the questions are yours to answer not the client's, there is never only one sub-question, and questions are open-ended rather than yes or no.
- Each question should be relevant, researchable, anchored, and precise, and a struggling technical design usually signals an inadequate question set.
- **Research framework**: a schematic of the steps needed to answer the questions, naming the knowledge required and the methods, whether borrowed or developed.
- **Operationalisation**: define key concepts exactly, state assumptions and what is neglected and why, then name the indicators or variables and how they are measured.

2.2.2 Explanation

The objective types are not interchangeable, and choosing among them is the first real commitment of the project. Theory-oriented work either builds a new theory or tests an existing relationship, while practice-oriented work runs along a spectrum from analysing a problem, through diagnosing and designing, to actively changing a situation and measuring the result. The objective template forces both halves of any objective into the open, since clause (a) states the contribution and clause (b) states the mechanism by which it is delivered. The four criteria then act as an acceptance test on that statement, asking whether a generic engineer would understand it, whether it benefits the problem, whether it fits the available resources, and whether it is precise about the contribution.

Research questions are the operational form of the objective, and the rules around them are practical guards rather than style points. The insistence on at least two sub-questions and on open-ended phrasing both push against the failure mode of a project that can be closed with a single yes or no, which would leave nothing to research. The relevant, researchable, anchored, and precise test mirrors the objective criteria at the level of each question. The framework then sequences the questions into a route, and the explicit invitation to reuse an existing method reflects the course view that a contribution can lie in the application or the combination of methods rather than in a new method itself. Operationalisation is the last conceptual act and the most consequential, because exact definitions, stated assumptions, named variables, and a measurement plan are precisely the inputs the technical design needs, and any vagueness left here surfaces later as a technical design that cannot be specified (section 2.3).

2.3 Part 3 — Technical Design: Strategy, Materials, and Planning

2.3.1 Content

- **Research strategy** takes the framework into practical terms and defines exactly how the research will be done.
- The strategy is judged on suitability of skills and problem level, availability of resources, achievable duration, and ethics.
- **Research materials** are every tool needed, from a computer with specific software to complex laboratory equipment.
- Secure access to materials as early as possible, allow time to learn the tools, and expect failure points with contingencies ready.
- **Research planning** sets the time frame for all activities, and because plans change it must allow flexibility.
- An initial plan includes milestones, deliverables, and holidays, and uses tools such as the Work Breakdown Structure, work flow diagram, and Gantt chart.

2.3.2 Explanation

The research strategy is where the theoretical framework first meets the constraints of the real world, and its four tests are a compressed feasibility study. Suitability asks whether the researcher and the problem are matched in skill and level, availability asks whether the needed resources can actually be obtained, duration asks whether the work fits the time, and ethics asks not only whether it can be done but whether it should. These are the same concerns Week 1 raised at the idea stage, now applied to a concrete plan rather than a topic. Materials are treated with unusual realism, since the guidance to secure access early, to budget learning time, and to plan for tools that fail is aimed squarely at the way technical projects actually slip.

Planning closes the technical design by attaching the work to a calendar, and its central message is that a plan is valuable precisely because it will change. Building in flexibility, naming milestones and deliverables, and even marking holidays are all ways of making the inevitable changes manageable rather than disruptive. The systems-engineering tools introduced here are named but not yet developed, because Week 5 returns to the Work Breakdown Structure and the Gantt chart and shows how they expose the critical path of a project (section 5.2). At this stage the point is simply that a schedule can only be assembled once the conceptual and technical designs are both in place, which is why planning sits last in the sequence.

2.4 Part 4 — Ethics and Research Conduct

2.4.1 Content

- Ensure the work is reproducible and can be traced back to original data, and keep the raw data.
- Verify the origin of any data that is given, and always give credit where it is due.
- Follow the university code of conduct as the governing standard for integrity.
- TU Delft operates the **Human Research Ethics Committee (HREC)** as its ethics board for research involving people.
- Permission must be sought for any research involving human test subjects, including questionnaire and interview participants and testers such as test pilots.

2.4.2 Explanation

The general recommendations look like good habits but are better read as the operational meaning of integrity in a technical project. Reproducibility and traceability to original data are what allow a result to be checked, reused, and trusted, and keeping the raw data is the precondition for both. Verifying the origin of supplied data and giving credit are the two failure points most likely to surface as misconduct, since unverified inputs can invalidate a whole analysis and missing credit is the core of plagiarism. These duties are not separate from the technical work, because they dictate how data is recorded and stored, which is exactly what Week 4 reproducibility and Week 5 data management formalise.

The Human Research Ethics Committee marks the one place in the course where ethical review is a hard procedural gate rather than a matter of personal care. Any study that touches human participants, whether through a questionnaire, an interview, or a physical test with a person, needs permission before it can proceed. This matters for planning, because approval takes time and must be sequenced into the schedule rather than assumed. It also connects forward to the personal-data rules of Week 3, where the same human participants generate data that falls under data-protection law (section 3.3), so the ethical gate and the legal gate are two views of the same obligation.

2.5 Reading Material

The Week 2 reading is the TU Delft Code of Conduct, which sits behind the brief conduct slides and gives the values those rules express. Where the slides give instructions, the code gives the reasoning, framing integrity as a shared institutional commitment rather than a list of prohibitions.

- The code is built on six core values: diversity, integrity, respect, engagement, courage, and trust.
- These values underpin the Integrity Statement, a set of basic principles every member of the community is expected to endorse and act upon.
- Integrity is framed as a collective responsibility, supported by an institutional integrity infrastructure of means, resources, and guidance.

- Its sections move from purpose and the core values, through acting with integrity and supporting integrity, to taking integrity further.

Chapter 3

Week 3 — Research Material and Data

Snapshot and part summaries: see the overview on page 7.

3.1 Part 1 — Population, Sample, and the Sampling Method

3.1.1 Content

- The **population** is the entire collection of things in the field, the entirety of the problem, and it is too large to study whole.
- The **sample** is the chosen subset of the population that is actually studied.
- A representative sample is a cross-section that allows generic statements over the population, and representativeness must be proven statistically with required sample sizes.
- If the sample is not representative, conclusions hold only for the studied sample and not for the whole population.
- The **sampling method** narrows in stages, beginning with the **theoretical population**, the group to which the study wants to generalise.
- It continues to the **study population**, the part that can actually be accessed, and the **sampling frame**, the means of listing and reaching them.
- It ends with the **sample**, the units that are actually in the study.

3.1.2 Explanation

The population and the sample are separated so that the scope of a conclusion is never left implicit. The population is the whole problem restated as a collection, which is by definition too large to study, so the sample is not a compromise but a necessity. What makes a sample useful is representativeness, and the course is firm that this is a statistical property to be demonstrated rather than assumed, complete with a required size. When representativeness fails, the work is not worthless, but its conclusions contract to cover only the sample studied, which is the disciplined form of the inductive caution from Week 1.

The four-stage funnel is the most practical idea in the part, because it shows that the sample is the survivor of several successive narrowings. The theoretical population is the ideal target of generalisation, the study population is the portion that is actually reachable, and the sampling frame is the concrete list or mechanism through which units are drawn. Each stage can lose coverage that the researcher never intended to drop, so the gap between the theoretical population and the final sample is where hidden bias accumulates. Naming the stages makes that gap visible and forces an honest statement of who the conclusions really describe.

3.2 Part 2 — Types of Samples and Standard Error

3.2.1 Content

- **Probability sampling** means every element has a known probability of selection.
- **Simple random sampling** draws so that every element is known and every possible sample has an equal chance of being selected.
- **Complex probability sampling** improves efficiency by dropping the equal-chance requirement while keeping known probabilities.
- Its forms include systematic random sampling, taking every nth element from a random start, and cluster random sampling, selecting whole groups then sampling within them.
- Multi-stage random sampling reiterates the sampling method within an already drawn sample.
- **Non-probability sampling** covers convenience and purposive sampling, neither of which can be assumed representative.
- **Convenience sampling** selects the sample because it is available, and this must be stated explicitly.
- **Purposive sampling** selects the sample because it holds desired information, for reasons such as expert advice, a needed quota, heterogeneity, or a snowball effect.
- The **standard error** is the precision of the statistical sample and is calculated from the **standard deviation**.
- The standard deviation is the spread of scores around their average in a single sample, the square root of the variance, in the same units as the measure.
- A greater standard deviation gives a greater error, and a greater sample size gives a smaller error.

3.2.2 Explanation

The division between probability and non-probability sampling is the operational version of the representativeness rule from the previous part. When every element has a known chance of selection, the mathematics of generalisation applies and the sample can stand in for the population, which is why simple and complex probability designs are the only ones that support statistical inference. Complex designs trade the simplicity of equal chances for efficiency, using structure such as clusters or stages to reach the same population at lower cost, while keeping the probabilities known. Non-probability designs abandon known probabilities entirely, so convenience and purposive sampling are explicitly barred from claiming to represent the population, and the course requires that this limitation be declared rather than hidden.

The standard error gives a number to the precision that representativeness only described in words. It is built from the standard deviation, which captures how spread out a single sample is, and it translates that spread into how reliably the sample estimates the population. The two stated levers are the whole practical content, since a larger spread inflates the error and a larger sample shrinks it toward the full-population value. This is the formal reason a study cannot buy precision for free, because driving the error down means collecting more data, which is exactly the cost and sample-size trade-off that sustainable research forces the researcher to justify in Week 4 (section 4.2).

3.3 Part 3 — Data Sources, Quality, and the Logbook

3.3.1 Content

- Data sources are someone else's data, theoretical data, simulated data, and experimental data from testing, surveying, or observing.
- **Personal data** is any data traceable directly or indirectly to a natural person, and it falls under the General Data Protection Regulation (GDPR).
- Commercially sensitive data arises in company theses, and no NDA or contract should be signed before the university gives legal advice.
- **Protected data** covers nuclear-weapons and missile-technology knowledge, which the UN and EU require be controlled through a knowledge-embargo exemption.
- **Qualitative data** is collected by observation of an environment, such as text, photos, and people.
- **Quantitative data** is the numerical result of measured or simulated variables, such as wind-tunnel data, finite-element results, and surveys.
- The data type follows from the research strategy, the research material, and the defined variables, and one type can sometimes be converted to the other.
- **Data quality** checks, run before analysis, look for typing errors, outliers, coding errors, and whether the data meets expectations.
- The **research logbook** records design choices, assumptions, how experiments were run, incidents, file locations, file names and changes, and people consulted.

3.3.2 Explanation

The four data sources line up with the four forms of measurement from Week 1, so the choice of source is really a continuation of the design decision made earlier. The legal categories then impose constraints that are independent of the science and cannot be negotiated away within the project. Personal data triggers the full weight of the GDPR and ties back to the human-subject gate of Week 2, commercially sensitive data turns a signature into a legal risk that must be cleared first, and protected data in fields such as hypersonics or re-entry requires an embargo exemption that takes real calendar time. Treating these as planning constraints rather than afterthoughts is the difference between a feasible project and one that stalls on paperwork.

The qualitative and quantitative split is presented as a consequence rather than a free choice, since the type of data is determined by the strategy, the materials, and the variables already fixed in the design. The note that data can be converted between the two, and that qualitative data can be summarised numerically, echoes the Week 1 point about attributes and keeps the boundary from being treated as rigid. Data quality is the gate immediately before analysis, and its checks exist because processing bad data produces confident wrong answers. The logbook is the instrument that makes all of this defensible, since a detailed record of choices, assumptions, incidents, and file histories is what allows the work to be reconstructed, which is the same traceability that ethics demands and that the data management plan of Week 5 formalises into a system.

3.4 Reading Material

The Week 3 reading is the TU Delft Data Usage and Storage Rules, which turn the brief data-rules slide into concrete obligations. Where the slide names the three categories, the reading specifies the procedures, the contacts, and the storage requirements that compliance actually demands.

- Personal data requires checking the HREC guidelines, collecting as little as possible, and taking special care with special categories such as race, religion, or health.
- Personal data must be stored in a secure database such as DANS and kept within the EU, since services like Google and Dropbox store data outside it and need extra measures.
- Personal data must be deleted or anonymised when no longer needed and anonymised as far as possible when publishing, with suspected leaks reported to the faculty data steward.
- Commercially sensitive data must follow the company's standards and be checked against the TU Delft open-access policy before anything is signed.
- Protected fields such as hypersonic aerodynamics, re-entry technology, guidance and navigation control, and launch and payload structures need a knowledge-embargo exemption that can take up to three months.

Chapter 4

Week 4 — Validation and Sustainability

Snapshot and part summaries: see the overview on page 9.

4.1 Part 1 — Verification and Validation

4.1.1 Content

- **Verification** is proving that the research method has been used correctly and is suitable for the research topic.
- **Validation** is proving that the outcomes are true and based on strong scientific evidence.
- Both are necessary to add credibility, to allow results to be used professionally, and to allow academic reproducibility and transparency.
- In industry, verification is a quality-control process checking compliance with regulations, specifications, or conditions set at the start of a phase.
- Industrial verification can occur in development, scale-up, or production, and it is often an internal process.
- In industry, validation is a quality-assurance process establishing high-assurance evidence that the system meets its intended requirements.
- Industrial validation involves acceptance of fitness for purpose with end users and stakeholders, and it is often an external process.
- In a thesis, ask how the methods will be verified, how the research goal can be validated and proven true, and how high the validation quality will be.
- Design a validation exercise into the work and reserve time to perform it rather than only exploring and implementing.

4.1.2 Explanation

Verification and validation answer two questions that are easy to confuse and dangerous to merge. Verification is about the method and asks whether it was applied correctly and whether it was the right method for the topic, which makes it an internal, process-facing check. Validation is about the outcome and asks whether the result is actually true and supported by strong evidence, which makes it an external, reality-facing check. The industrial framing sharpens the split, with verification as quality control against fixed specifications and validation as quality assurance of fitness for purpose with real users. A project can pass one and fail the other, and a verified method that produces a plausible but false result is exactly the failure validation exists to catch.

The thesis guidance turns the distinction into a planning instruction rather than a definition to memorise. Asking in advance how methods will be verified and how the goal will be validated forces both checks into the schedule, and the explicit warning to reserve time for validation reflects how often it is squeezed out by exploration and implementation. This is why verification and validation belong with planning from Week 2, since a validation exercise that was never timetabled tends not to happen. Both are also the operational meaning of the credibility and reproducibility that the ethics of Week 2 and the sustainability of the next part demand, so they sit at the centre of what makes a result usable beyond the project that produced it.

4.2 Part 2 — Engineering for One Planet

4.2.1 Content

- The **Engineering for One Planet (EOP)** framework asks that planning consider environmental, social, cultural, and economic implications.
- It covers three aspects: the value of research, the reproducibility of research, and resource management.
- **Value of research** requires forecasting both near-term and long-term cost and value, not only immediate findings.
- Its questions ask whether there is potential value to science or industry, whether the work can be built upon, and whether it carries direct or unintended potential for harm.
- **Reproducibility of research** signals transparency and rigour and lets later work explore similar topics with fewer resources.
- Its barriers are the complexity of the work, technology change, human error, and intellectual property rights.
- Good data management overcomes the technology-change and human-error barriers in particular.
- **Resource management** is awareness of the resources used, split into digital resources and methodologies.
- Digital guidance favours local work over servers, local storage over cloud, downloading documents accessed often, printing only what will be read at least three times, and limiting extra screens.
- Methodology guidance asks whether the desired accuracy is realistic for the resources, whether a less intensive alternative exists, and whether the sample size is justified.

4.2.2 Explanation

The EOP framework recasts sustainability as something decided during design rather than judged after the fact, and its three axes each attach to choices made elsewhere in the course. The value axis is a forecasting duty, asking the researcher to weigh long-term benefit and potential harm before committing, which extends the gap-finding of the Week 1 literature review into a judgement about whether the gap should be filled at all. The reproducibility axis restates the credibility argument of verification and validation in resource terms, since reproducible work can be reused by others without repeating the full cost. Its four barriers are worth holding because two of them, technology change and human error, are explicitly the ones that disciplined data management is said to solve, which is why this axis points directly at Week 5.

Resource management is where the framework becomes concrete, and it deliberately spans both the digital habits of daily work and the methodological choice of how to gather evidence. The digital recommendations are quantitative rules of thumb about where energy is spent, from server use to cloud storage to printing, and their shared logic is to move work and data to the least costly location that still serves the purpose. The methodology questions are the more consequential half, because they put a sustainability test on the accuracy a study aims for, asking whether the target is realistic, whether a lighter setup would do, and whether the sample size is justified. This is the same trade-off the standard error described in Week 3, now given an explicit reason to keep the sample no larger than the objective requires, since every extra test or iteration spends real resources.

4.3 Reading Material

> The Week 4 reading, Conducting Sustainable Research, expands the EOP slides into a working method for making a project sustainable. It connects the value axis to literature-gap analysis and turns the reproducibility axis into a concrete set of data-management habits.

- It ties the value of research to identifying a literature gap and judging that gap on its environmental, social, cultural, and economic implications.
- It recommends scripted analysis software over graphical tools to remove human error and make every repetition identical.
- It asks for descriptive, self-explanatory file and variable names carrying version dates and numbers, stored in non-proprietary open formats such as text and csv.
- It advises keeping an uncorrected original data file and documenting the workflow from raw data to results so an outsider can reproduce them.
- It extends the server guidance to troubleshooting locally, compressing and limiting transferred files, and deleting server files once the work is done.
- It frames methodology as a balance between resource use and quality, captured by garbage in equals garbage out, with sample size justified against the objective.

Chapter 5

Week 5 — Data Management and Project Planning

Snapshot and part summaries: see the overview on page 10.

5.1 Part 1 — The Data Management Plan

5.1.1 Content

- The **data management plan (DMP)** must be designed before collection, because data is very hard to organise only at the end.
- Good data management stores raw data together with metadata and the analysis workflow, so the data can be understood and reused in the long term.
- Section I, **Data Description**, lists the data type, file formats, how data is collected with source and terms of use, the purpose of processing, the storage location, and who has access.
- Section II, **Documentation and Data Quality**, names the accompanying documentation: collection methodology, an electronic lab notebook, a README file, and a data dictionary of variables.
- Section III, **Storage and Backup**, fixes where data and code are stored and applies the **3-2-1 rule**: three copies, on two storage media, with one copy in a different physical location.
- Section IV, **Legal and Ethical Requirements**, asks about confidential or classified data and about human subjects or third-party datasets from human participants.
- Section V, **Data Sharing and Long-term Preservation**, sets what is shared, how, for example through 4TU.ResearchData, and under which license such as CC0, CC-BY, CC BY-SA, or CC BY-NC.
- Section VI, **Data Management Responsibilities**, asks whether TU Delft leads the project and who becomes responsible for the data if you leave.
- Further recommendations: use descriptive file and variable names, keep a consistent filing system, keep an uncorrected original data file, prefer open formats, and document the workflow.

5.1.2 Explanation

The data management plan is the point where good intentions about data become an enforced structure, and its defining feature is that it is written first. The course is explicit that organising data at the end is close to impossible, so the plan is a precondition of collection rather than a report on it. The six sections walk through the entire life of a dataset, from describing what it is, through documenting and storing it, to clearing its legal status, sharing it, and assigning responsibility for it after the project ends. Reading them as six questions makes the plan portable, since any project can be interrogated with the same list regardless of its field.

Several sections deserve particular attention because they encode rules rather than prompts. The 3-2-1 rule in the storage section is a memorable backup standard, with three copies on two kinds of media and one copy kept physically elsewhere, and it is meant to be designed in from the start rather than improvised after a loss. The sharing section introduces licensing as a deliberate choice, ranging from the public-domain CC0 to the more restrictive non-commercial license, which determines how freely the data can be reused. The responsibilities section forces a question students rarely consider, namely who owns the data once they graduate, which is exactly the long-term stewardship that the Week 4 reproducibility argument and the reading's FAIR principles are concerned with.

5.2 Part 2 — Project Management in Research

5.2.1 Content

- Project management gives a research project organised progress, resource optimisation, early risk mitigation, and effective communication with the supervisory team.
- **Estimating tasks** starts by outlining every task, then estimating each, using comparative, parametric, or bottom-up estimating.
- The **Work Breakdown Structure (WBS)** divides the project into work packages, tasks, and sometimes sub-tasks.
- The WBS is taken to a level of detail that allows a time estimate per task, and the individual estimates are summed to the project total.
- **Resource management** treats resources as people, equipment, and facilities, some reusable like people and some consumable like a plate of alloy.
- Most resources are not always available and require tasks to gain access, so each task's required resources and their availability must be described.
- **Project planning** is the time-based articulation of the plan and can only be assembled once the conceptual and technical design are outlined.
- The **Gantt chart** is the common way to show the plan, and it reveals the **critical path**, the chain that determines the shortest time to complete the project.

5.2.2 Explanation

Project management is imported into research to solve a specific problem, that a research plan without structure tends to slip, and its four stated benefits are the symptoms it treats. Estimation comes first because nothing can be scheduled until its duration is known, and the three estimating methods offer different routes to that number, from comparison with past work to parametric models to summing the pieces. The Work Breakdown Structure is the tool that makes estimation possible, since decomposing the project into work packages and tasks continues until each task is small enough to estimate honestly, and the total then emerges by summing upward. This is the disciplined version of the planning step introduced in Week 2, now with a method for producing the numbers a plan needs.

Resource management adds the constraint that tasks do not run on time alone, since they consume people, equipment, and facilities that are not always available. The distinction between reusable and consumable resources matters for scheduling, because a reusable resource like a person becomes a shared bottleneck across tasks while a consumable one is spent. Planning then assembles these estimates and constraints into a time-based schedule, and the course is firm that this is only possible once both the conceptual and technical designs are complete, which is the same ordering rule seen throughout the course. The Gantt chart is the standard representation, and its real value is exposing the critical path, the longest dependent chain of tasks that fixes the minimum project duration, since that chain is where any delay directly lengthens the whole project.

5.3 Reading Material

The Week 5 reading, Sustainable Data Management by Pommerening, supplies the principles behind the plan, most importantly the FAIR framework and a categorisation of research data. It explains why documentation and workflow capture are the difference between data that lasts and data that becomes unusable.

- It defines **FAIR principles** as findability, accessibility, interoperability, and reuse, with reuse as the ultimate objective that good documentation serves.
- It classifies research data into five categories: observational, experimental, simulation, derived, and metadata.
- Observational data is tied to time and place and irreplaceable, while experimental data is reproducible but often expensive to regenerate.
- It notes that data outlives the project that created it, so preserving and sharing data is now part of the scientific process rather than an afterthought.
- It warns that spreadsheet-driven analysis quickly becomes impossible to reconstruct, so an automated scripted workflow is easier to repeat and revise.
- It stresses that missing documentation of metadata and workflow leaves no reproducibility, no transparency, and no possibility of reuse.

Chapter 6

Week 6 — Reporting and Writing

Snapshot and part summaries: see the overview on page 11.

6.1 Part 1 — Technical Reports

6.1.1 Content

- Communicating the work is part of the job rather than an addition to it, and the goal is to let a reader understand a subject or carry out a task.
- Industry reports include project results, handover documentation, instruction manuals, and software guides.
- Academic reports include assignments, the master or doctoral thesis, conference articles, and journal articles.
- A technical report conveys specific information about a technical subject to a specific audience for a specific and practical purpose.
- The **Research Methodologies report** conveys your research plan about your topic to a technical person familiar with your field, in order to organise your research activities.
- A **report structure** has a head, the introduction that orients the reader and stays proportional to the body, a main body of connected sections, and feet, the conclusions on which the report stands and which add no new information.
- The **Research Methodologies report structure** is an introduction on why the topic matters, a body of research questions, methods with tools and expected results, and planning, and conclusions that summarise the whole plan.
- Write little and often, favouring snack writing over binge writing, and alternate thinking and writing rather than separating them.

6.1.2 Explanation

The part opens by reframing reporting as part of the engineering work itself, not a chore appended to it, because a result that is not communicated cannot be used. The definition of a technical report is built entirely from specifics, a specific subject, audience, and purpose, which is what separates it from general writing and what makes the audience the first thing to fix. The Research Methodologies report is then a special case of that definition, with the plan as the subject, a knowledgeable peer as the audience, and the organising of research activities as the purpose. Naming the purpose this way matters, because it explains why the report is structured around a plan rather than around results that do not yet exist.

The skeleton metaphor gives the structure its logic, since the introduction is proportioned to support the body, the body carries the argument in connected sections, and the conclusions are load-bearing feet that introduce nothing new. Mapping this onto the Research Methodologies report shows how directly it draws on the course, with the body's research questions coming from Week 2 (section 2.2) and its planning coming from Week 5 (section 5.2). The writing-habit advice is practical rather than stylistic, since alternating thinking and writing treats writing as a way to clarify thought rather than a transcription of finished ideas. This is why the course recommends frequent short sessions, because a report assembled in one late burst tends to expose exactly the gaps that steady writing would have surfaced early.

6.2 Part 2 — Referencing

6.2.1 Content

- Use at least three different document types, including at least one book for theoretical background.
- Use at most two websites, or no more than 10% of the list, and include patents, standards, and technical documents where relevant.
- Most sources should be peer-reviewed journal articles, and most should be less than ten years old.
- Choose one reference style and keep to it throughout.
- The **author-date style (APA)** cites parenthetically as (Author et al., 2020) or narratively as Author (2020), with the list ordered alphabetically by author.
- Its advantage is that a reader recognises important original texts from the in-text reference, and its disadvantage is that longer in-text references reduce readability.
- The **numbered style (AIAA)** cites as a bracketed number such as [1], with the list ordered numerically by order of appearance.
- Its advantage is easy movement from one reference to the next, and its disadvantage is that the in-text number carries no information itself.
- The **reference checklist** requires every borrowed illustration and statement to be referenced, with quotations in quotation marks and paraphrases in your own words.
- The list must contain all references used in the text and only those references used in the text.

6.2.2 Explanation

The sourcing rules are quality controls on the evidence base of the report, and each one targets a specific weakness. Requiring several document types and at least one book pushes against a literature base built entirely from convenient web pages, while capping websites and favouring peer-reviewed articles raises the average reliability of the sources. The currency rule, that most sources be under ten years old, guards against a report resting on superseded work, which connects back to the literature-review habits of Week 1 (section 1.4). These are not arbitrary quotas but proxies for a literature base that a knowledgeable reader would trust.

The two citation styles are presented as a genuine trade-off rather than a matter of taste, and understanding the trade-off is what makes the choice defensible. Author-date places the source identity directly in the text, so a reader recognises a landmark paper without consulting the list, at the cost of interrupting the prose with names and dates. Numbered style keeps the text clean and lets a reader walk the references in order, at the cost of an in-text marker that says nothing on its own. The firm instruction to pick one and keep it reflects that the damage comes from inconsistency, not from either choice, and the reference checklist then operationalises the give-credit duty by demanding that the text and the list agree exactly in both directions.

6.3 Part 3 — Writing Style

6.3.1 Content

- The reader is the most important person, so first define the target reader, what they want, and what they need the information for.
- Then guide them in, remembering that readers look first at titles, then at figures, then at conclusions, and only then at the rest.
- Layout must look clean with well-placed high-resolution images, and block diagrams often convey more than detailed explanations.
- The **4Cs style principles** are presented as the course's named set of writing principles.
- Graph tips: use tick marks with major gridlines at the start, often the origin, and the end, and add a non-overlapping legend when there is more than one line.
- Keep markers the same colour as their line, use straight or short-dashed lines rather than dotted or long-dashed, and keep units and fonts consistent.
- **AI tools for scientific writing** must be used responsibly, since text generation can become redundant, brainstorming only looks back in time, and generated content or references are prone to error.
- The **final writing checklist** covers spelling and grammar, one or two ideas per paragraph, consistent format and nomenclature, and acronyms defined on first use.
- It also requires every figure to be readable in PDF and captioned, clear without the text, and a title and conclusions that stand on their own.

6.3.2 Explanation

The organising idea of the part is that the reader, not the writer, decides whether a report succeeds, which inverts the usual instinct to write for oneself. Defining the target reader first is what makes every later choice answerable, since the right level of detail and the right emphasis depend entirely on who is reading and why. The observation that readers scan titles, then figures, then conclusions before the body is the practical reason those elements must be self-contained, because they are read in isolation and often instead of the full text. This is also why figures and block diagrams are treated as primary carriers of meaning rather than decoration, and it extends the Week 3 concern with clear, well-labelled data presentation.

The guidance on AI tools is notably cautious and worth reading as a set of limitations rather than an endorsement. Each named use has a specific failure mode, with generation tending to overwrite, brainstorming bounded by a backward-looking training horizon, and content or reference generation prone to outright error, so the tools assist but do not absolve the writer of responsibility. The graph conventions are concrete because small inconsistencies in ticks, legends, line styles, and units are exactly what make a figure hard to read at a glance. The final checklist then closes the course by acting as a single gate over the whole report, and its recurring theme, that titles, figures, and conclusions must be understandable on their own, is the same reader-centred principle applied one last time.

Chapter 7

Cross-Week Concept Threads

The course is taught week by week, but it is built as two threads that each run the length of the programme. This chapter follows both threads end to end, showing how the output of one week becomes the input of the next. It is the chapter to read when a question spans weeks rather than sitting inside one.

7.1 The Conceptual Track

The conceptual thread begins in Week 1 with the language of research and the shape of its reasoning. The vocabulary fixed there, the hypothesis, the variable, and the three study types (section 1.2), is what later weeks manipulate, and the hourglass logic of deduction down to measurement and induction back to generalisation (section 1.3) is the inferential frame the whole track inherits. Nothing here is yet a plan, but everything the plan will be made of is named, which is why the track starts with definitions rather than activities.

Week 2 turns that vocabulary into a design. The hypothesis becomes a research objective, the objective generates research questions, and the questions are sequenced by the research framework (section 2.2). The decisive step is operationalisation (section 2.2), where the abstract concepts of Week 1 are pinned to exact definitions, stated assumptions, and named variables, because this is the point at which the conceptual design hands concrete quantities to the technical design (section 2.3). The split between the conceptual question of what and why and the technical question of how and when (section 2.1) is the organising move of the week, and it only works because the language of Week 1 made the concepts precise enough to operationalise.

Week 6 closes the thread by writing it down. The operationalised plan becomes the body of the Research Methodologies report, whose sections are the research questions and the planning built earlier (section 6.1), and the literature that seeded the objective in Week 1 (section 1.4) returns as the reference list that must meet the sourcing and style rules of Week 6 (section 6.2). Read in sequence, the conceptual track is a single sentence stretched across the course: a precise question, made into a designed plan, then communicated to a reader who can act on it.

7.2 The Empirical Track

The empirical thread begins in Week 1 with the six steps of a research project (section 1.1), and two of those steps carry the whole track. The sample step asks which portion of the problem will be studied, and the measure step asks how observations will be recorded, and both are stated only in outline at this stage. The track is the progressive sharpening of these two steps into rigorous, checkable, and managed practice over the following weeks.

Week 3 makes the sample and measure steps exact. The sample step becomes the population-to-sample funnel and the statistics of representativeness (section 3.1), with the probability boundary deciding what can be generalised and the standard error quantifying precision (section 3.2). The measure step becomes the treatment of data sources, the legal categories that constrain them, the qualitative and quantitative split, and the quality checks and logbook that protect them (section 3.3). What was a one-word step in Week 1 is now a method with rules and a record.

Week 4 then interrogates the data the empirical track has produced. Verification asks whether the method was applied correctly and validation asks whether the result is true (section 4.1), and sustainable research adds a third interrogation, weighing the value, reproducibility, and resource cost of the work through the Engineering for One Planet framework (section 4.2). The resource axis in particular reaches back to the standard error of Week 3, since it demands that the sample be no larger than the objective justifies. Week 5 finally manages both the data and the project, as the logbook of Week 3 grows into the six-section data management plan (section 5.1) and the planning outlined in Week 2 grows into a Work Breakdown Structure and a Gantt chart with a critical path (section 5.2). The empirical track is therefore a tightening loop, from a rough step, to a rigorous method, to a checked result, to a managed system, with each week formalising what the previous one left informal.

7.3 Where the Tracks Meet

The two tracks are not parallel lines that never touch. Sustainable research sits on the empirical side yet feeds project management and data management, so the value and resource questions of one track shape the planning decisions of the other (section 4.2). Data management collects the output of data, sustainable research, and technical design at once (section 5.1), which is why the course map draws it as the node where the red and green chains converge. The Research Methodologies report is the final meeting point, since its body carries the conceptual plan while its conduct, sourcing, and data handling carry the empirical discipline, and the report guide that follows (chapter 8) shows section by section how both tracks are written into one document.

Chapter 8

Report Writing Guide

The Research Methodologies report is where the whole course is written into a single document. Its structure was given in Week 6 as a head, a body, and feet (section 6.1), and each part of that structure is generated by concepts taught in specific earlier weeks. This chapter maps every report section to its source in the course and then notes what the section requires and where it commonly goes wrong.

8.1 Section-to-Concept Map

Table 8.1 traces each section of the report back to the week and concept that produces its content. Reading down the table is the fastest way to see which part of the course to revisit when a section of the report will not come together.

Report section	Generated by	What it must contain
Introduction (head)	Week 1 problem and research ideas (section 1.1, section 1.4); value of research (section 4.2)	Why the topic matters, the problem it addresses, the gap in the literature, and the relevance to science or industry, kept proportional to the body.
Research Questions	Week 2 objective and questions (section 2.2)	The research objective and at least two open-ended sub-questions that are relevant, researchable, anchored, and precise.
Methods, Tools, and Expected Results	Week 2 technical design (section 2.3); Week 3 sampling and data (section 3.1, section 3.3); Week 4 verification and validation (section 4.1)	The strategy and materials, the sampling and data approach, the variables and how they are measured, and how methods and outcomes will be verified and validated.
Planning	Week 2 planning (section 2.3); Week 5 project management and data management (section 5.2, section 5.1)	A Work Breakdown Structure, a schedule with milestones and a critical path, the resources required, and the data management approach.
Conclusions (feet)	Week 6 report structure (section 6.1)	A concise summary of the entire plan, understandable on its own, introducing no new information.

Table 8.1: Each report section and the course week and concept that generate its content.

8.2 Section Notes and Common Pitfalls

The introduction is the head of the report, and its job is to make a knowledgeable reader care about the topic before the plan is presented. It draws on the problem formulation and idea sources of Week 1 and on the value judgement of the Engineering for One Planet framework, so it should state the gap in the literature and the relevance of closing it. The common pitfall is disproportion, where an introduction grows longer than the body it is meant to support, which the skeleton metaphor of Week 6 explicitly warns against. A second pitfall is asserting importance without grounding it in the literature, which leaves the reader with a claim rather than a reason.

The research questions section is the spine of the body and comes directly from the conceptual design of Week 2. It must present the objective and at least two open-ended sub-questions that pass the relevant, researchable, anchored, and precise test, because a single closed question leaves nothing to investigate. The most frequent error is importing the client's questions instead of the researcher's own, which produces questions the project is not designed to answer. A related error is questions so vague that the methods section cannot be specified, which is the signal Week 2 gives that the question set itself needs rework.

The methods, tools, and expected results section is where the technical design and the empirical track are written down together. It must name the strategy and materials from Week 2, the sampling and data approach from Week 3, and the verification and validation plan from Week 4, and it must connect each method to the variables it produces. The usual pitfall is describing methods in isolation from the questions they serve, so a reader cannot see why each method is present. A second pitfall is omitting validation entirely, treating it as something to add later, which the course warns leaves no time reserved for it.

The planning section turns the plan into a schedule and draws on both Week 2 and Week 5. It should present a Work Breakdown Structure decomposed to estimable tasks, a Gantt chart that exposes the critical path, the resources each task needs, and the data management approach that will keep the work reproducible. The common pitfall is a schedule with no critical path, which hides where delay actually costs time, and a second is a plan with no flexibility, which the course notes will not survive contact with the project. The conclusions are the feet of the report, summarising the entire plan so completely that a reader could grasp it from the conclusions alone, and their defining pitfall is introducing new information that belongs in the body.

Appendix A

Checklists

These checklists collect the criteria and rules scattered through the week chapters into a single place for use during the exam. Each box names the section it is drawn from. The square boxes are for ticking against your own work.

Research Objective Criteria (section 2.2)

- ☐ **Informative**: could someone with generic engineering knowledge understand what the project is about?
- ☐ **Useful**: what is the benefit of the research to the problem or solution?
- ☐ **Feasible**: is it possible within your capabilities and resources?
- ☐ **Clear**: is it precise about the project's contribution?

Research Question Formulation (section 2.2)

- ☐ The questions are yours to answer, not the client's.
- ☐ There are at least two sub-questions, never only one.
- ☐ Questions are open-ended rather than yes or no.
- ☐ Each question is relevant, researchable, anchored, and precise.
- ☐ The question names the key concept and links to quantitative evidence.
- ☐ The set builds toward a testable main hypothesis.
- ☐ If the technical design struggles, the questions are revisited and edited.

Data Quality Checks (section 3.3)

- ☐ Checked for typing errors.
- ☐ Checked for outliers.
- ☐ Checked for errors in coding the data.
- ☐ Checked whether the data meets expectations.
- ☐ A detailed research logbook is kept.
- ☐ A data management plan is in place.

Data Management Plan Sections (section 5.1)

- ☐ **I. Data description**: types, formats, collection, purpose, storage, access, and volume.
- ☐ **II. Documentation and data quality**: methodology, lab notebook, README, and data dictionary.
- ☐ **III. Storage and backup**: storage location and the 3-2-1 rule of three copies, two media, one off-site.
- ☐ **IV. Legal and ethical requirements**: confidential data, human subjects, and third-party datasets.
- ☐ **V. Sharing and long-term preservation**: what is shared, how, and under which license.
- ☐ **VI. Responsibilities**: lead institution and who owns the data after you leave.

Reference Checklist (section 6.2)

- ☐ All borrowed illustrations are referenced.
- ☐ All borrowed statements are substantiated with references.
- ☐ Direct citations are in quotation marks and paraphrases are in your own words.
- ☐ Each citation or paraphrase has a correct in-text reference in the chosen style.
- ☐ All references are correct and complete in the chosen style.
- ☐ The list contains all references used in the text.
- ☐ The list contains only references used in the text.

Final Writing Checklist (section 6.3)

- ☐ No spelling or grammar errors.
- ☐ Each paragraph conveys one or two ideas.
- ☐ The paragraph format is consistent throughout.
- ☐ The nomenclature is consistent throughout.
- ☐ All acronyms are defined on first use.
- ☐ All figures are readable in PDF.
- ☐ All figures and tables include a caption.
- ☐ Figures are clear without reading the text.
- ☐ The title captures the contents.
- ☐ The introduction captures the ideal reader.
- ☐ Conclusions are understood without reading the rest.

Sustainable Research Reflection (section 4.2)

- ☐ Is there potential value to science or industry from the findings?
- ☐ Can the research be built upon?
- ☐ Is there direct potential harm to the environment, society, or economy as intended?
- ☐ Could the research be used for unintended harmful purposes?
- ☐ Is the desired accuracy realistic given the resources available?
- ☐ Is there a less resource-intensive alternative with reasonable accuracy?
- ☐ Is the sample size justified?

Appendix B

Glossary

Each entry gives a one-line definition and a pointer to the section where the term is treated in full. For page-level lookup of any bold term in the text, use the index at the very back of the guide.

3-2-1 rule A backup standard of three copies of the data, on two different storage media, with one copy in a separate physical location (see section 5.1).

AIAA style (numbered) A citation style that marks sources with bracketed numbers in order of appearance and lists them numerically, keeping the text clean but uninformative in itself (see section 6.2).

APA style (author-date) A citation style that names author and year in the text and lists sources alphabetically, letting a reader recognise key works at the cost of longer in-text references (see section 6.2).

Attribute A value that a variable can take, which may be quantitative or qualitative and must be exhaustive and mutually exclusive (see section 1.2).

Causal study A study designed to determine whether one or more variables cause or affect one or more outcome variables (see section 1.2).

Complex probability sampling Probability sampling that improves efficiency by dropping the equal-chance requirement while keeping selection probabilities known, as in systematic, cluster, and multi-stage designs (see section 3.2).

Conceptual research design The strategic area that models the content of the research and fixes what, why, and how much is studied (see section 2.1).

Convenience sampling Non-probability sampling in which units are chosen because they are available, which must be stated and cannot be assumed representative (see section 3.2).

Correlation versus causation The principle that an observed association between variables does not establish that one drives the other, since a third variable may drive both (see section 1.2).

Critical path The longest chain of dependent tasks in a schedule, which sets the shortest possible duration of the whole project (see section 5.2).

Data management plan (DMP) A six-section plan, designed before collection, that describes, documents, stores, clears, shares, and assigns responsibility for a project's data (see section 5.1).

Data quality The set of checks run before analysis for typing errors, outliers, coding errors, and agreement with expectations (see section 3.3).

Deduction Reasoning that moves from the more general to the more specific, forming the narrowing half of the hourglass (see section 1.3).

Descriptive study A study designed to document the values of one or more unknown variables (see section 1.2).

Engineering for One Planet (EOP) A framework that judges research on its value, its reproducibility, and its resource management across environmental, social, cultural, and economic implications (see section 4.2).

FAIR principles The guiding principles that data should be findable, accessible, interoperable, and reusable, with reuse as the ultimate objective (see section 5.1).

Feasibility The trade-off between rigour and practicalities, covering duration, ethics, cooperation, cost, and access, that gates whether an idea becomes a project (see section 1.4).

Gantt chart A time-based chart of a project's tasks that schedules the work and exposes the critical path (see section 5.2).

Hourglass model The shape of a single study, narrowing from broad questions through operationalisation and measurement, then widening back to generalised conclusions (see section 1.3).

Human Research Ethics Committee (HREC) The TU Delft board whose permission is required for any research involving human test subjects (see section 2.4).

Hypothesis A specific statement of prediction describing in concrete terms what a study expects to happen, whose form fixes the study type (see section 1.2).

Induction Reasoning that moves from specific observations to broader generalisations, the fallible widening half of the hourglass (see section 1.3).

Literature review A survey that identifies related research and sets the project within a conceptual and theoretical context, locating the gap to be filled (see section 1.4).

Operationalisation The conceptual step that turns abstract concepts into exact definitions, stated assumptions, named variables, and a measurement plan (see section 2.2).

Personal data Any data traceable directly or indirectly to a natural person, which falls under the General Data Protection Regulation (see section 3.3).

Population The entire collection of things in a field, the whole problem restated as a collection and too large to study directly (see section 3.1).

Probability sampling Sampling in which every element has a known probability of selection, the precondition for statistical generalisation (see section 3.2).

Protected data Knowledge in fields such as nuclear weapons and missile technology, subject to a knowledge embargo requiring an exemption (see section 3.3).

Purposive sampling Non-probability sampling in which units are chosen because they hold desired information, which must be stated and cannot be assumed representative (see section 3.2).

Qualitative data Data collected by observation of an environment, such as text, photographs, and people (see section 3.3).

Quantitative data The numerical result of measured or simulated variables, such as wind-tunnel data, finite-element results, and surveys (see section 3.3).

Relational study A study investigating the relationship between two or more variables (see section 1.2).

Reproducibility The property that work can be repeated by others, signalling rigour and enabling reuse with fewer resources (see section 4.2).

Research framework A schematic representation of the steps needed to answer the research questions, naming the required knowledge and methods (see section 2.2).

Research logbook A detailed record of design choices, assumptions, procedures, incidents, file histories, and people consulted that makes a project traceable (see section 3.3).

Research objective A statement of the contribution a project makes to a problem and the way that contribution is delivered, judged informative, useful, feasible, and clear (see section 2.2).

Research question An open-ended question, one of at least two, that yields information needed for the objective and enables the technical design (see section 2.2).

Research strategy The technical-design phase that defines exactly how the research will be done, judged on suitability, availability, duration, and ethics (see section 2.3).

Resource management Awareness and control of the people, equipment, facilities, and digital resources a project consumes, split into digital resources and methodologies (see section 4.2).

Sample The chosen subset of a population that is actually studied (see section 3.1).

Sampling frame The concrete means of listing and accessing the study population from which the sample is drawn (see section 3.1).

Simple random sampling Drawing so that every element is known and every possible sample has an equal probability of selection (see section 3.2).

Standard deviation The spread of scores around their average in a single sample, the square root of the variance, in the same units as the measure (see section 3.2).

Standard error The precision of a statistical sample, calculated from the standard deviation, falling as the sample grows and the deviation shrinks (see section 3.2).

Study population The part of the theoretical population that can actually be accessed (see section 3.1).

Technical report A document that conveys specific information about a technical subject to a specific audience for a specific and practical purpose (see section 6.1).

Technical research design The strategic area that plans the execution of the research and fixes how, where, and when through strategy, materials, and planning (see section 2.1).

Theoretical population The ideal group to which a study wants to generalise, the starting point of the sampling funnel (see section 3.1).

Validation Proving that the outcomes of the research are true and based on strong scientific evidence, an outcome-facing and often external check (see section 4.1).

Value of research The EOP axis that requires forecasting the near-term and long-term benefit and potential harm of the work (see section 4.2).

Variable The smallest block in the language of research, whose possible values are its attributes (see section 1.2).

Verification Proving that the research method has been used correctly and is suitable for the topic, a process-facing and often internal check (see section 4.1).

Work Breakdown Structure (WBS) A decomposition of a project into work packages and tasks, taken to a level of detail that allows each task to be estimated (see section 5.2).

Index